

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Medium

Question

In the xy -plane, line s passes through the point $(0, 0)$ and is parallel to the line represented by the equation $y = 18x + 2$. If line s also passes through the point $(4, d)$, what is the value of d ?

Answer

- A. 2
- B. 18
- C. 72
- D. 74

Correct Answer: C

Rationale

Choice C is correct. A line in the xy -plane can be represented by an equation of the form $y = mx + b$, where m is the slope and b is the y -coordinate of the y -intercept of the line. It's given that line s passes through the point $(0, 0)$. Therefore, the y -coordinate of the y -intercept of line s is 0. It's also given that line s is parallel to the line represented by the equation $y = 18x + 2$. Since parallel lines have the same slope, it follows that the slope of line s is 18. Therefore, line s can be represented by the equation $y = mx + b$, where $m = 18$ and $b = 0$. Substituting 18 for m and 0 for b in $y = mx + b$ yields the equation $y = 18x + 0$, or $y = 18x$. If line s passes through the point $(4, d)$, then when $x = 4$, $y = d$ for the equation $y = 18x$. Substituting 4 for x and d for y in this equation yields $d = 18(4)$, or $d = 72$.

Choice A is incorrect. This is the y -coordinate of the y -intercept of the line represented by the equation $y = 18x + 2$.

Choice B is incorrect. This is the slope of the line represented by the equation $y = 18x + 2$.

Choice D is incorrect. The line represented by the equation $y = 18x + 2$, not line s , passes through the point $(4, 74)$.